# USB4 2.0 ENGINEERING NOTICE REQUEST FORM

## Title: DP Tunneling tDPAUXtoWAKE
## Applied to: USB4 Specification Version 2.0

| **Brief description of the functional changes** |
|---|
| Enlarge the time it requires the DP IN to send a set config wake by 5 uS |

| **Benefits as a result of the changes:** |
|---|
| Allows easier support by DP IN Adapter design such as complete previous packet handling |

| **An assessment of the impact to the existing revision and systems that currently conform to the USB specification:** |
|---|
| None |

| **An analysis of the hardware implications:** |
|---|
| None |

| **An analysis of the software implications:** |
|---|
| None |

| **An analysis of the compliance testing implications:** |
|---|
| None |

# USB4 2.0 ENGINEERING NOTICE REQUEST FORM

## Actual Change

### (a). Section 10.4.14 DP PHY Testability

## 10.9 Timing Parameters

Table 10-28 lists the timing parameters for a DP Adapter.

**Table 10-28. DP Adapter Timing Parameters**

| Parameter | Description | Min | Max | Units |
|---|---|---|---|---|
| tDPAUXtoWAKE | The maximum time between a DP IN Adapter receiving the end of the AUX_SYNC pattern of an AUX Request over the DP AUX and sending the SET_CONFIG Packet of type AUX_PATH_CLX. | | ~~5~~10 | µs |

For reference:

#### 10.4.15.1.2 CLx Exit

If a DP IN Adapter initiates an AUX Path sleep handshake and does not send a SET_CONFIG Packet over the AUX Path after completing the handshake, then upon reception of an AUX Request, the DP IN Adapter shall send a SET_CONFIG Packet of type AUX_PATH_CLX with the *S/W* bit set to 0b. The following rules determine when the SET_CONFIG Packet shall be sent:

- If the DP IN Adapter sent a PM Packet with the *CLx State* field set to 01b as part of the last AUX Path sleep handshake, then it shall send the SET_CONFIG Packet within tDPAUXtoWAKE after receiving the end of the AUX_SYNC pattern of the AUX Request.
- Else it shall send the SET_CONFIG Packet within tDPAUXDelayIn after receiving the complete AUX Request.